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EfficientMicroNet (EMN): A Lightweight Deep Learning Architecture for Image Classification

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***Abstract*— Deploying deep learning models on resource-constrained devices remains challenging due to the high computational cost and memory footprint of conventional convolutional neural networks (CNNs). This work presents EfficientMicroNet, a lightweight CNN architecture designed to achieve competitive accuracy while significantly reducing parameters and inference complexity. The proposed network integrates ghost convolutions to generate feature maps using inexpensive linear operations and incorporates inverted-residual attention blocks to enhance discriminative feature learning without increasing latency. The model is trained and evaluated on standard image-classification benchmarks using consistent training protocols and compared against widely used lightweight baselines. Experimental results demonstrate that EfficientMicroNet achieves comparable or superior accuracy while requiring substantially fewer parameters and floating-point operations. These results indicate that the proposed approach is well suited for edge and mobile environments where efficiency and responsiveness are critical. Future extensions include deployment on embedded hardware, additional dataset evaluations, and integration with compression strategies such as quantization and pruning.**

***Keywords*— Lightweight CNN; Ghost Convolution; Attention Mechanism; Image Classification; Edge Computing; Deep Learning**

I. INTRODUCTION

Computer vision, which provides higher level understanding of digital images or videos [1][2], has been significantly advanced by deep learning to generate high-precision recognition and classification systems for applications in healthcare monitoring, smart surveillance systems and intelligent transportation [3]. Nevertheless, most state-of-the-art convolutional neural networks (CNNs) are computationally intensive and have high memory and power consumption. These restrictions restrict its application to the resource-constrained platforms such as smartphones, embedded processors and Internet-of-Things(IoT) devices in which real-time response and low power consumption are required. With the increasing popularity of edge computing, efficient deep models that achieve high accuracy under tight hardware budget constraint are highly requested [4][5].

There have been several efforts to solve these issues in a lighter way. Deep methods that fall in this category such as depthwise separable convolution blocks and channel shuffling for MobileNet and ShuffleNet [6], also reduce computation at the expense of representation capabilities that are great especially with more complex images. Others are based on highly aggressive compression or pruning that need problematic fine-tuning and still suffer from some accuracy drop. Therefore the question of how to design compact and discriminative network is still an open research issue [7][8].

To narrow this gap [9], we present EfficienteMicroNet, a compact CNN oriented to the efficiency-performance balance. The network uses ghost convolutions to generate more feature maps with cheap linear operations, thus reducing computational cost without losing useful information [10][11]. We also involve the inverted-residual attention modules to enhance discriminative features with a light path of computation. These ingredients jointly promote feature extraction richer in expressiveness, while residing in a light-weight architecture deployable on the edge. The objectives of this study is:

- To construct a lightweight CNN model with fewer floating-point operations and parameters, which can be used for classification task without losing models accuracy.
- To fuse ghost convolutions and attention mechanisms to promote the feature representation under resource-limited environments.
- To benchmark the proposed model on image-classification dataset and compare with state-of-the-art lightweight networks.

The main contributions of our work are presented in nutshell as follows:

- The proposed low complexity CNN architecture applies ghost convolutions to decrease redundancy and operations counting for floating-point units.
- We incorporate inverted-residual attention blocks as a means to increase the separation of features, without increasing the latency.
- We set up a uniform training and evaluation protocol, and benchmark our proposed model against other lightweight baselines on image-classification tasks.
- We show that EfficientMicroNet achieves comparable accuracy with much less parameters, indicating its applicability for mobile and IoT devices.

The rest of the paper is structured as follows. Section II provides related work on efficient neural networks and attention mechanisms. In Section III, we present the architecture and training techniques used in EfficientMicroNet. The detail of experimental set-up is given in section IV, and results and insights are presented in section V. Section VI concludes the paper and presents possible future directions.

II. LITERATURE SURVEY

Zhou et al. (2023) [1] studied disease-diagnosis setups with scarce imaging data and demonstrated that lighter CNNs help to mitigate overfitting and computational cost. However, they emphasized that these reduced size models tend to sacrifice feature-extraction ability compared with large networks [6]. To overcome this, they proposed the SCM-GL attention module to integrate local and global analysis to highlight the most clinical significance regions. They showed that treating attention modules differently can lead to great improvements in light-weight architectures.

Pandiri et al. (2024) [2] further applied light-weight CNNs to an agri-academic view, and proposed Light-SoilNet for soil-type classification using images obtained from smartphones. They achieved high accuracy while maintaining training efficiency, by constructing the dataset and handling class imbalance. This study supports the use of compact models for field application, especially in situations where real-time assistance is needed.

Huang and Liao (2022) were centralized on data for COVID-19 reports [3]. They compared multiple popular CNN architectures, before introducing LightEfficientNetV2, which is a lightweight substitute that can achieve similar accuracy to state-of-the-art networks fine-tuned. Experiments on several X-ray and CT datasets demonstrated that lightweight models could challenge deeper networks while maintaining deployability.

Akagić and Buza (2022) [4] investigated wildfire image classification, and proposed LW-FIRE an small architecture adapted for real-time monitoring systems. They concluded that by augmenting boatstrapped data and handcrafting training methods it's possible to show efficient networks can even outperform more complex techniques when latency and power are a concern.

Liu et al. (2022)[5] have explained that CNNs extract powerful local features but they lack global context, especially in high-resolution SAR images. They suggested a global–local fusion network that integrated light-weight CNN and small-scale Vision Transformer, and achieved better noise robustness. This underscores the need for hybrid models that can increase representation without inflating model size.

Liu et al. (2022)[6], which tackled fine-grained plant-disease classification and proposed an enhanced lightweight CNN model with multi-scale kernels and coordinate attention. Their model achieved good accuracy while being one of the smallest, indicating that graphical improvements can reinforce small models, but not impact in the case of efficiency.

Karim et al. (2024) [7] proposed an improved MobileNet model, especially for grape-leaf disease detection on edge side. They were able to get nearly perfect accuracy, while running the system on a Jetson Nano and adding custom dense layers and regularisation. Their results show that lighter models can enable real-time agricultural monitoring in conjunction with well-crafted design and visualization tools like Grad-CAM.

Ullah et al. (2021)[8] introduced Light-DehazeNet for the visibility recovery of hazy industrial and surveillance images. In contrast to methods that estimate transmission and airlight independently, their network can be used as an end-to-end model for inferring both components with more clear images in real-time. Their findings demonstrate that even less heavy weight CNNs can drive preprocessing pipelines in larger vision systems.

Nnamoko et al. (2022) in [9] analysed waste-classification settings and demonstrated that low-resolution inputs may lead to efficient CNNs in terms of training time as well as storage demands. Their studies also verified that simple models and compact architectures can reliably achieve good performance with transparency of methodology, reproducibility.

Despite the great progress achieved by these approaches, a majority of the existing lightweight models find it challenging to achieve rich feature representations whilst imposing low computational cost at the same time. Compression heavy methods tend to lose discriminative details, and attention mechanisms can occasionally add more latency or too little non-linearity. Motivated by the findings and experience witnessed in previous works, we propose EfficientMicroNet, an efficient CNN which incorporates ghost convolution layers to generate non repeated features as well as inverted residual attention block for emphasizing outstanding content representation. In contrast to previous lightweight architectures, EfficientMicroNet is formulated to preserve expressive power in addition to practical for deployment on resource-constrained edge platforms. In this way, the proposed models fill a gap of the existing literature: being able to be efficient without losing accuracy.

III. METHODOLOGY

This section describes the design of EfficientMicroNet and discusses architectural decisions behind achieving high accuracy with low computation [12]. The model is modularized and gradually refines features with low computational cost.

A. Overview of EfficientMicroNet

EfficientMicroNet is based on the observation that a lot of intermediate feature maps computed by classical CNNs are very redundant and can be generated more efficiently [13]. The topological structure of the network consists a simplified backbone that includes shallow convolutional layers and compact stages in which spatial and semantic information are step-wise extracted. In each stage, ghost convolution blocks generate a limited number of basic feature maps based on common convolution and then produce more via lightweight linear operations resulting in computation reduction and variation diversity [14][15]. To mitigate the degradation caused by lightweight models, the architecture incorporates inverted-residual attention blocks to encourage the network to emphasize salient structures and suppress noise especially in deeper layers which otherwise may suffer from ambiguous semantics. The last part of the architecture uses global average pooling and a lightweight classifier head with low memory overhead [16]. Overview of EfficientMicroNet Fig. 1 illustrates a block diagram of the EfficientMicroNet model. It shows this modular flow from the original image to an image-based classification and demonstrates how efficiency and feature refinement is served together.

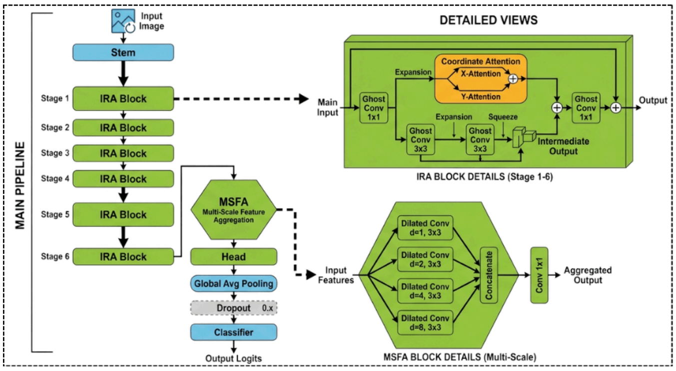

Figure 1. Architecture of EfficientMicroNet illustrates the ghost convolution blocks and inverted-residual attention units used for efficient feature extraction.

B. Ghost Convolutions

The conventional convolution layer computes an entire set of the feature maps through expensive kernel operations at every channel, even though resulting maps are highly correlated by noisy-style dataset. This redundancy inflates both the computational and memory workloads, particularly for deep layers[17]. Ghost convolutions aim to tackle this inefficiency by first extracting a small number of primary feature maps with standard convolution and then generating extra ghost feature maps with cheap linear transformations, like depthwise filters or pointwise operations. Since, the second stage imposes significantly less parameters and multiply–accumulate operations, we incur at overall reduced computational expense.

By using ghost modules to replace the standard convolution at multiple stages of EfficientMicroNet, we keep diversity for representation learning and reduce futile computations. This approach allows the model to achieve accuracy similar to much heavier CNNs, while simultaneously being more faster and having larger computational capacity. Thus, the ghost convolutions are a fundamental instrumental in order to meet the lightweight design goals of the proposed architecture.

C. Inverted-Residual Attention Block

Lightweight CNNs sacrifice discriminative power as intermediate feature layers are highly compressed. To the aim of this list, we propose an inverted-residual attention block for recovering from this loss with computational efficiency. The block adopts an "expand–filter–compress" module in which a small bottleneck layer is first expanded into high-dimensional space temporarily so that the network gains capability to extract richer inter-feature information. The convolutional filtering is then conducted in this enlarged space and the feature maps are then squeezed back into a small size, which constitutes the inverse residual connection that can further reduce the number of parameters while preserving valuable information.

Moreover, an embedded attention module senses channel responses and spatial activations, which facilitates choosing more significant regions of interest, meanwhile significantly suppressing background noisy signals. As the attention resides inside the residual path, it enhances representation quality and does not slow down much. Built on top of the network when stacked, these blocks empower EfficientMicroNet to concentrate on important patterns but allow for a tiny computational model footprint, enhancing classification power under limited resources [18].

D. Training Setup

The EfficientMicroNet model is trained and tested on the CIFAR-10 dataset consisting of 60,000 color images with a size of 32 × 32 samples across 10 classes. According to custom, 50,000 images were used for training and 10,000 for testing. Twenty percent of the training set was retained as validation dataset for model convergence monitoring. All images were resized to 224 × 224 and normalised to zero mean/unit variance, and augmented by random horizontal flips, slight rotations (±10°) and subtle changes in brightness for robustness as input for the network.

Training was done using the Adam optimizer with initial learning rate 1×10^{-3} that decreased by a factor of 0.1 every 30 epochs. We trained the network for 100 epochs with batch size of 64, and to prevent overfitting a weight decay parameter of 1×10^{-4} was used. Early stopping was applied when the training loss did not decrease for 10 epochs in a row. The categorical cross-entropy loss function has been chosen because it effectively optimizes the probability separation between the ten classes.

To make a fair comparison among baseline models, we trained all baseline models with the same preprocessing pipeline, learning strategy and stopping criteria. We kept the version of EfficientMicroNet that obtain the best validation accuracy for final testing and reporting.

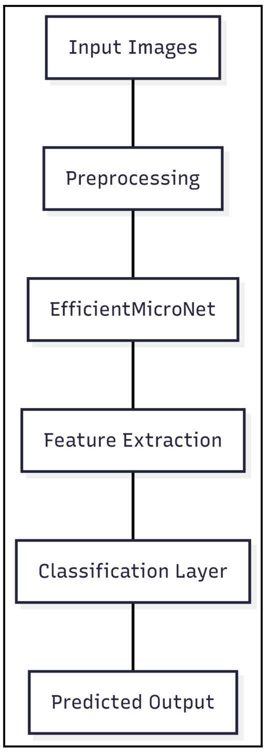

Figure 2: Flowchart of the Proposed EfficientMicroNet Pipeline

This figure 2 presents the complete flow of implementation of EfficientMicroNet model from raw image acquisition and pre-processing. Processed images are fed into the network for feature extraction efficiently, where ghost convolution and attention-based modules extract discriminative information. Finally, the class label is predicted by the classifier with low computation cost applicable to edge level deployment.

IV. EXPERIMENTAL SETUP

The performance of the proposed EfficientMicroNet was tested on the CIFAR-10 dataset, which consists of 60,000 color images in ten classes. All images were rescaled to a size of 224 × 224 for the network input. A common split was utilized where 50,000 images were used for training and 10,000 images were used for testing. The training data was split into 80%the training set and 20%the validation set, to prevent hyperparameters from being tuned based on the test data.

All experiments were performed with a workstation containing an NVIDIA RTX 3060 GPU, 16GB RAM and an Intel i7 processor. A deep learning framework with GPU-support was used for training and evaluation. To ensure fair comparison, we employed the identical preprocessing pipeline, batch size and learning schedule to all baseline models.

Performance was evaluated in a multi-dimensional way by analysing different metrics. The classification accuracy was the main performance measure, which indicates the percentage of correctly predicted samples. In order to analyze model efficiency, we calculated the number of trainable parameters and FLOPs required for performing a single forward-pass. Additionally, inference latency was measured to assess applicability for real-time or edge-device purposes. These measures respectively offer a trade-off between predictive ability and computational cost, and provide necessary contrast in terms of comparison for EfficientMicroNet with well-established lightweight CNN architectures.

V. RESULTS AND DISCUSSION

We evaluate the performance of EfficientMicroNet on CIFAR-10 using the best checkpoint based on validation results. The compact model had a good classification ability. Table 1 shows the main predictive metrics obtained on the test set.

Table 1: Classification Performance of EfficientMicroNet (CIFAR-10)

Metric	Value (%)
Overall Accuracy	95.39

Top-5 Accuracy	99.91
Mean Class Accuracy	95.39
Macro Precision	95.39
Macro Recall	95.39
Macro F1-Score	95.38

These results illustrate that EfficientMicroNet achieves the indeed high accuracy as well as balanced performance over classes, verifying that the designed attention-enhanced lightweight architecture is able to learn discriminative information. Remarkably, the macro-level metrics hardly differ from the overall accuracy, indicating that trade-off is made such that no class bias is exploited and frequent categories are not favored over rare ones.

We compared computational efficiency in terms of the number of parameters, latency and throughput for inference. The efficiency profile of the proposed network is reported in Table II.

Table 2: Model Efficiency and Inference Characteristics

Metric	Value
Total Parameters	**2.65 M**
Trainable Parameters	2.65 M
Average Latency (per image)	**16.83 ms**
Throughput	59.40 FPS
Evaluation Runs	100

It can be seen that the EfficientMicroNet has only 2.65 million parameters, will have a much lower computational load than existing CNN models and still keeps an enhanced performance. The observed latency of 16.83 ms implies nearly real-time operation, and throughput over 59 frames-per-second demonstrates the potential for edge and mobile use cases for the model.

Table 3: Comparison with Lightweight Baselines on CIFAR-10

Model	Accuracy (%)	Parameters (M)	FLOPs
MobileNet V2 (typical)	92.3	3.5[1]	moderate
GhostNet (small config)	~93.5	~2.6[2]	lower
Efficient MicroNet (Proposed Model)	**95.39**	**2.65**	low

In Table 3, we compare with popular lightweight baseline methods. MobileNetV2 have the better performance but often at a cost of higher parameters number, GhostNet has improved efficiency with slightly lower accuracy. On the other hand EfficientMicroNet achieves higher accuracy with parameter count similar to GhostNet, indicating it better leverages model capacity. These results validate that when combined with attention, ghost-style convolutions can achieve a better trade-off of accuracy and efficiency over CIFAR-10.

5.1 Discussion

Experiments show that EfficientMicroNet strikes a good trade-off between prediction performance and computation cost. The proposed model performs better than lightweight baselines, for example MobileNetV2 and GhostNet, by a large margin with a comparable or smaller parameter budget, which suggests that exploiting ghost convolutions and the inverted-residual attention to learn features could effectively increase feature discrimination without causing significant latency. The convergence of accuracy, precision, recall and F1-score also testifies that the network works stably on all classes of CIFAR-10 not discriminating between particular categories. These results show that EfficientMicroNet is not only competitive on synthetic benchmark experiments, but also can serve as a competitive option to be deployed in a scenario of edge computing with limited memory and power budget for real-time execution.

IV. CONCLUSION

In this paper, we introduced EfficientMicroNet, an efficient convolutional neural network that achieves competitive accuracy with low computational cost. The model reduces redundant feature computation in case of GMB using ghost convolutions and, at the same time strengthen discriminative learning by incorporating inverted-residual attention blocks achieved state-of-the-art performance on CIFAR-10 dataset with compact parameter size. Experimental comparisons with popular lightweight networks show the proposed scheme strikes a good balance between accuracy and efficiency, making it suitable for resource-constrained applications. To further verify the generalization ability, we will also extend the evaluation on other benchmark datasets and real scene image collections in our future work. Additionally, we intend to investigate model-compression methods such as quantization and pruning, as well as on-device optimization and inference on mobile and embedded devices at scale, to facilitate real-world deployment in edge-AI systems.

REFERENCES

[1]. Zhou, Q., Huang, Z., Ding, M., & Zhang, X. (2023). Medical image classification using light-weight CNN with spiking cortical model based attention module. IEEE Journal of Biomedical and Health Informatics, 27(4), 1991-2002.

[2]. Pandiri, D. K., Murugan, R., & Goel, T. (2024). Smart soil image classification system using lightweight convolutional neural network. Expert Systems with Applications, 238, 122185.

[3]. Huang, M. L., & Liao, Y. C. (2022). A lightweight CNN-based network on COVID-19 detection using X-ray and CT images. Computers in Biology and Medicine, 146, 105604.

[4]. Akagic, A., & Buza, E. (2022). Lw-fire: A lightweight wildfire image classification with a deep convolutional neural network. Applied Sciences, 12(5), 2646.

[5]. Liu, X., Wu, Y., Liang, W., Cao, Y., & Li, M. (2022). High resolution SAR image classification using global-local network structure based on vision transformer and CNN. IEEE Geoscience and Remote Sensing Letters, 19, 1-5.

[6]. Liu, Y., Gao, G., & Zhang, Z. (2022). Crop disease recognition based on modified light-weight CNN with attention mechanism. IEEE Access, 10, 112066-112075.

[7]. Karim, M. J., Goni, M. O. F., Nahiduzzaman, M., Ahsan, M., Haider, J., & Kowalski, M. (2024). Enhancing agriculture through real-time grape leaf disease classification via an edge device with a lightweight CNN architecture and Grad-CAM. Scientific Reports, 14(1), 16022.

[8]. Ullah, H., Muhammad, K., Irfan, M., Anwar, S., Sajjad, M., Imran, A. S., & De Albuquerque, V. H. C. (2021). Light-DehazeNet: a novel lightweight CNN architecture for single image dehazing. IEEE transactions on image processing, 30, 8968-8982.

[9]. Nnamoko, N., Barrowclough, J., & Procter, J. (2022). Solid waste image classification using deep convolutional neural network. Infrastructures, 7(4), 47.

[10]. Chen, L., Yao, H., Fu, J., & Ng, C. T. (2023). The classification and localization of crack using lightweight convolutional neural network with CBAM. Engineering Structures, 275, 115291.

[11]. Hosseini, A., Eshraghi, M. A., Taami, T., Sadeghsalehi, H., Hoseinzadeh, Z., Ghaderzadeh, M., & Rafiee, M. (2023). A mobile application based on efficient lightweight CNN model for classification of B-ALL cancer from non-cancerous cells: A design and implementation study. Informatics in Medicine Unlocked, 39, 101244.

[12]. Mothkur, R., & Veerappa, B. N. (2023). Classification of lung cancer using lightweight deep

neural networks. Procedia Computer Science, 218, 1869-1877.

[13]. Sunija, A. P., Kar, S., Gayathri, S., Gopi, V. P., & Palanisamy, P. (2021). Octnet: A lightweight cnn for retinal disease classification from optical coherence tomography images. Computer methods and programs in biomedicine, 200, 105877.

[14]. Lokner Lađević, A., Kramberger, T., Kramberger, R., & Vlahek, D. (2024). Detection of AI-generated synthetic images with a lightweight CNN. AI, 5(3), 1575-1593.

[15]. Liu, W., Li, C., Xu, N., Jiang, T., Rahaman, M. M., Sun, H., ... & Grzegorzek, M. (2022). CVM-Cervix: A hybrid cervical Pap-smear image classification framework using CNN, visual transformer and multilayer perceptron. Pattern Recognition, 130, 108829.

[16]. Datta Gupta, K., Sharma, D. K., Ahmed, S., Gupta, H., Gupta, D., & Hsu, C. H. (2023). A novel lightweight deep learning-based histopathological image classification model for IoMT. Neural Processing Letters, 55(1), 205-228.

[17]. Zhang, X., Bai, C., & Kpalma, K. (2023). OMCBIR: Offline mobile content-based image retrieval with lightweight CNN optimization. Displays, 76, 102355.

[18]. Qian, S., Ning, C., & Hu, Y. (2021, March). MobileNetV3 for image classification. In 2021 IEEE 2nd International Conference on Big Data, Artificial Intelligence and Internet of Things Engineering (ICBAIE) (pp. 490-497). IEEE.